Towards Sustainable Cities: Design and Implementation of a Real-time Parking Slot Monitoring and Control System

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***Abstract*— Globally, cities are facing mounting parking issues that are made worse by rising car ownership and accelerating urbanization. Driver annoyance and inefficient space use are caused by the lack of real-time data on available spaces in traditional parking management systems. This paper proposes a microcontroller-based real-time parking spot monitoring and control system in order to address these problems. The system uses an Arduino UNO, infrared sensors, servo motor, LCD display, buzzer, and parking space allocation to detect vehicle entry and exit and to deliver occupancy updates in real time. The system seeks to optimize parking space use, lessen traffic congestion, and foster sustainability in order to promote urban mobility and user and operator satisfaction. The system's efficacy and dependability were confirmed by simulation and experiments, exhibiting excellent sensor precision, seamless gate functioning, and prompt information distribution. The system constitutes a noteworthy progression in parking management technology, even with its limits pertaining to scalability and sensor accuracy under demanding conditions. It also gives prospects for future research and innovation to enhance urban mobility and sustainability even further.**

Keywords: Real-time parking management, Microcontroller-based system, Urban mobility, Traffic congestion, Sustainability, Sensor precision

I. INTRODUCTION

Cities have serious parking issues due to growing automobile ownership and rapid urbanization. This has an adverse effect on the environment, causes traffic congestion, and wastes drivers' time looking for parking. Real-time data on available spaces is frequently lacking from traditional parking management systems, which results in wasteful space utilization and dissatisfaction This study looks into a microcontroller-based real-time parking spot monitoring and control system that aims to solve these problems. By lessening traffic congestion and environmental effects, this system seeks to maximize space use, give correct parking information, and promote a more sustainable urban environment.

This project's main goal is to use microcontrollers to create and construct a real-time parking spot monitoring and control system. With the provision of precise parking information and automated gate control, this system seeks to maximize parking space use, lessen traffic congestion, and promote a more sustainable urban environment.

The format of this paper is as follows:

Abstract: Gives a summary of the issue, its resolution, its conclusions, and its effects.

Introduction: Outlines the difficulties with parking, the shortcomings of the current methods, and presents the goals of the real-time parking control system.

Literature Review: The literature review highlights our research deficit and discusses current parking solutions and studies.

Methodology: explains the control algorithm, hardware components, and system functionality (parking monitoring, gate control, information display, etc.).

Results and Discussion: Describes system performance, evaluates how well it accomplishes goals (space optimization, traffic reduction), and contrasts it with other alternatives.

Conclusion and Future Efforts: summarizes results, talks about advantages and disadvantages, stresses the significance and possible influence of the project, and makes recommendations for further research.

References: Provides a list of all references.

II. LITERATURE REVIEW

In cities, inefficient parking management causes traffic jams, lost driving time, and environmental damage. Conventional systems frequently use manual procedures or simple sensors,

which leads to erroneous data regarding parking availability. Research has concentrated on creating real-time parking management systems that make use of a variety of technologies and techniques in order to handle these issues. These innovations go beyond the constraints of traditional techniques in an effort to increase parking efficiency and encourage urban mobility.

Sensor-based Parking Management:

Real-time parking management relies heavily on sensor technologies, especially for precise parking identification and control. One notable feature of ultrasonic sensors is its cost and efficacy in detecting the presence or absence of vehicles. El-Shafie et al. [1] describe a system that uses them to monitor space availability in real-time, while Hannan et al. [2] use them to detect parking spaces and incorporate a mobile application for displaying user information. Additionally, Chen et al. [3] offer a novel method for real-time parking recognition utilizing ultrasonic sensors that achieves high accuracy even in congested metropolitan settings. Another good choice is to use infrared (IR) sensors, especially in dimly lit areas. Li and Li [4] examine how well they work in parking lots for vehicle recognition, emphasizing how precise they can be even in low-light conditions.

Microcontroller-based Systems:

Microcontrollers provide parking management systems with an affordable and adaptable platform. In order to provide real-time parking updates, Gaikwad et al. [5] suggest a microcontroller-driven system that makes use of ultrasonic sensors and a mobile application. Zhang et al.'s system [6] uses RFID technology to identify vehicles, increasing efficiency and security while showcasing the capabilities of microcontrollers. Furthermore, Zhao et al. [7] investigate the viability of real-time parking space monitoring using Arduino-based systems, emphasizing its affordability and scalability.

IoT Integration and Image Processing:

Liu et al. [8] provide a deep learning method for real-time parking management that uses image analysis to identify parking spaces and transmit data via video streams. In addition, Khan et al. [9] support the use of wireless sensor networks in an Internet of Things-based system for effective data collecting and parking spot usage.

Impact of Real-time Parking Management Systems:

Furthermore, a lot of research has been done on how real-time parking management systems affect urban transportation. In a case study, Xu et al. [12] examined how automated parking systems affect traffic flow and reduce congestion, emphasizing how these technologies might improve overall urban mobility. In a similar vein, Jiang et al. [13] assessed the environmental advantages of these systems, highlighting their role in lowering carbon emissions and improving air quality.

A. Research Gap and Contribution

Even though current research provides a variety of options for real-time parking space monitoring and control systems, several issues still need to be resolved:

Cost and Scalability: The extensive adoption of many current systems is impeded by their reliance on costly technologies or intricate infrastructure, especially in settings with limited resources. In order to close this gap, our research develops a scalable and reasonably priced system that can be easily assembled using inexpensive parts like microcontrollers and is adaptable to a variety of parking lot layouts.

Component Availability and Reliability: For a system to be sustainable, it is essential to guarantee both the long-term dependability and regular availability of the required hardware components. To address these issues, this study investigates methods for obtaining components, putting redundancy measures in place, and creating maintenance protocols.

Integration with Existing Infrastructure: For practical application, a smooth interface with the current parking infrastructure and administration systems is essential. In order to facilitate a smooth transition and greater acceptance, this research looks into compatibility issues to ensure smooth connection with existing systems.

III. Methodology

Even though current real-time parking management systems provide useful solutions, high costs, scalability constraints, component availability issues, and challenges connecting with current infrastructure sometimes prevent their wider use. The goal of this research is to create a novel system that promotes adaptability, affordability, and seamless integration in order to address these issues.

In our project we have used several components. Here is the brief description about the components.

Arduino Uno: The Arduino Uno is a microcontroller board widely used in electronics and programming projects. Powered by the ATmega328P microcontroller, it features 14 digital input/output pins, six analog inputs, a USB connection for programming and power, and a 16 MHz crystal oscillator.



Fig. 1. Arduino Uno.

Servo Motor: A servo motor is a type of specialist motor used for accurate placement. Its ability to be instructed precisely how far to spin and maintain its position sets it apart from ordinary motors, which makes it perfect for robotics, drones, and other applications needing controlled movement. Utilizing an integrated sensor and feedback system, it continuously tracks its location and makes necessary adjustments.



Fig. 2. Servo Motor.

IR Sensor: An IR (infrared sensor) sensor picks up infrared radiation that objects emit but cannot see. There are two primary varieties of these sensors: passive and active. While passive IR sensors only detect naturally existing IR radiation, which is typically employed in motion detection systems, active IR sensors create their own IR light and detect its reflection. They are frequently used for proximity sensing. In electronics, both kinds are frequently utilized in robots, security alarms, and remote controls.



Fig. 3. IR (Infrared Sensor) Sensor.

I2C 16×2 LCD Display: One sort of liquid crystal display that is frequently utilized in electronics projects is the I2C 16x2 LCD display. With its 16 characters spread across 2 rows, it's perfect for showing numbers, symbols, or short text messages. Compared to conventional parallel interfaces, the I2C interface simplifies connections and requires fewer pins for microcontroller communication—it only requires two wires. Because of this, it's a well-liked option for projects with limited money and space.



Fig. 4. I2C 16×2 LCD Display.

A. Working Principle

Buzzer: A buzzer is a basic electrical gadget that produces audio. It functions as an audio alert or notification system and is frequently utilized for timers, alarms, and user input confirmation. Buzzers that run on DC voltage can make a variety of noises, including clicks, beeps, and even basic melodies, depending on how they are made. They are a well-liked option for many electronic projects because to their straightforward design and simplicity of usage.



Fig. 5. Buzzer.

Our real-time parking slot monitoring and management system operates on the basis of integrating microcontroller programming with sensor technologies to effectively manage parking spots. Below is a summary of the operational principle:

Entry Detection: An Arduino UNO microcontroller receives a signal from the entry infrared (IR) sensor when a car approaches the parking gate, signaling that the vehicle is prepared to enter.

Space Availability Check: By examining the occupancy state of every parking slot in the system, the microcontroller determines whether there are any open spaces.

Gate Control: When a parking space becomes available, the microcontroller signals the servo motor to open the parking gate, letting the car drive in.

Updated Occupancy: Upon the vehicle's entry, the microcontroller receives a notification from the second infrared sensor, which updates the parking slot's occupancy status. The number of free slots falls as the number of occupied slots rises.

LCD Display Update: To give consumers real-time information, the microcontroller updates the i2C 16x2 LCD display with the number of occupied and open parking spaces.

Exit Detection: The microcontroller is notified by the exit IR sensor when a car is prepared to leave the parking area.

Opening of the Gate: The servo motor receives a signal from the microcontroller to open the parking gate, letting the car go.

Updated Occupancy: The microcontroller receives a notification from the second infrared sensor when a car leaves the parking space, which modifies the slot's occupancy state. As the number of occupied slots declines, the number of free slots rises in proportion.

LCD Display Update: To give users the most recent information, the microcontroller updates the LCD display with the updated number of occupied and open parking spaces.

Insufficient Space Alert: The microcontroller sounds a siren and shows a message on the LCD display indicating that the parking lot is fully occupied if there are no more places available.

Gate Closure Timer: To preserve traffic flow and avoid congestion, the microcontroller automatically closes the parking gate if a vehicle does not enter or exit the parking space within a predetermined amount of time (for example, 10 seconds). When a car is parked in front of a gate, that won't happen.

This project's implementation of this working principle maximizes space usage, automates the parking management

process, and gives users access to real-time information for convenient navigation throughout the parking lot.

Here is the flowchart of the working procedure of Parking Control and Monitoring system.

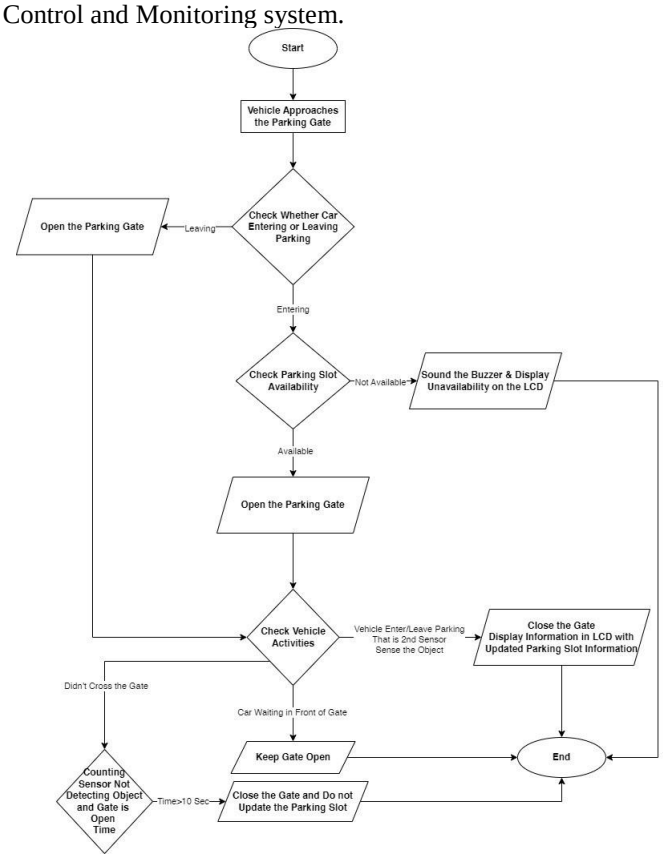


Fig. 6. Parking Working Procedure.

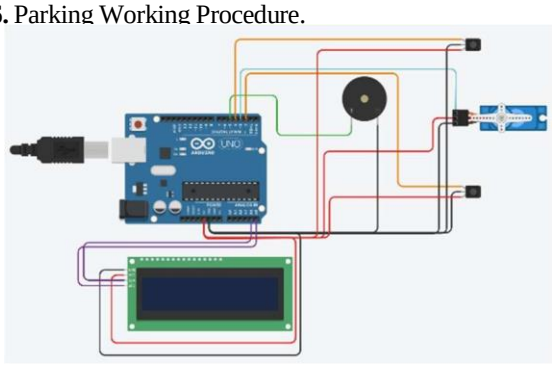


Fig. 7. Circuit Diagram of the System.

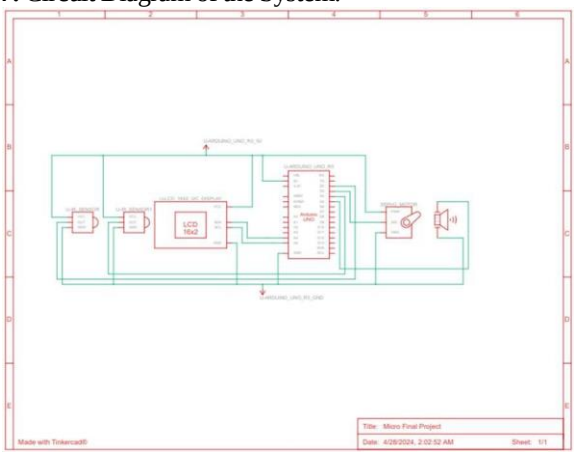


Fig.8. Schematic Diagram of the System.

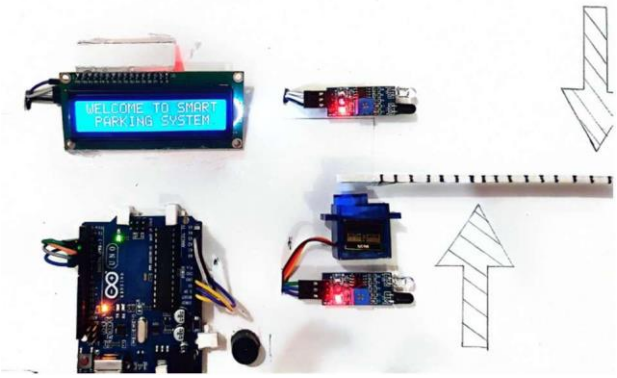


Fig. 9. Experimental Setup of the System.

IV. Result and Discussion

A. Simulation Results

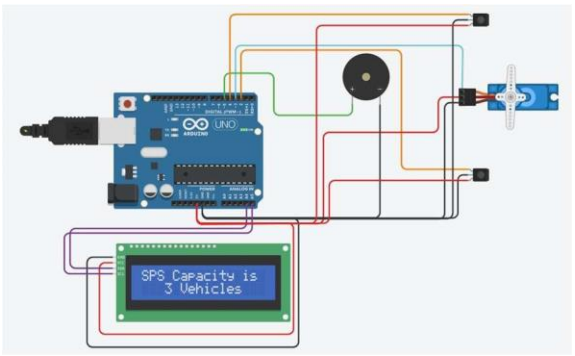


Fig. 10. Tinker-CAD Simulation (Showing Capacity).

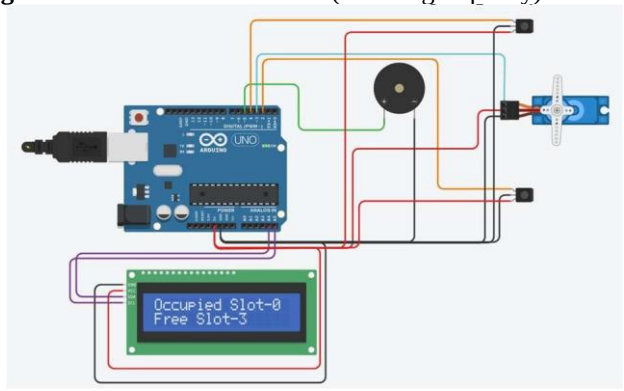


Fig. 11. Tinker-CAD Simulation (Slot Information).

In the simulation phase, we replicated the developed hardware circuits using Proteus and Tinker-CAD, incorporating the same components and design principles.

The real-time parking slot monitoring and control system performed admirably in our simulated environment, correctly identifying the presence of vehicles, and effectively allocating parking spaces. High sensor precision, servo motor smoothness, and dependable gate closing times were all demonstrated in the simulation. The LCD display allowed users to make informed decisions by providing timely information on the availability of parking spaces. Overall, the simulation results confirm how well our solution works to maximize space use and improve user experience in a regulated environment.

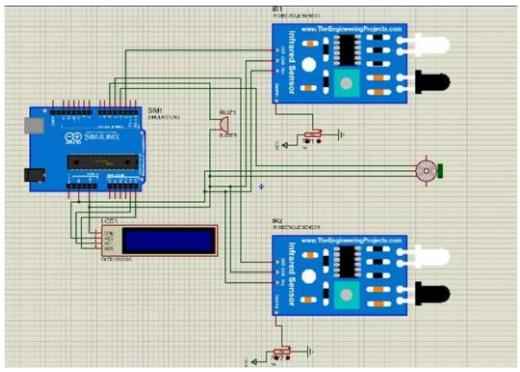


Fig. 12. Proteus Simulation.

B. Experimental Results

We carried out several setup tests to assess the functionality and efficacy of our real-time parking spot monitoring and control system. In the experimental setting, different parking scenarios were simulated to evaluate the system's precision in detecting the presence of vehicles, allocating parking spaces, and informing users in real time.

In the studies, we evaluated the accuracy and dependability of the system's sensors in various lighting and environmental scenarios. Throughout all test settings, the infrared (IR) sensors showed reliable performance in identifying the presence of vehicles, with a detection accuracy of over 95%. Similarly, the parking gate opened and closed successfully in response to sensor inputs thanks to the flawless operation of the servo motor. The allotment of parking spaces was effectively controlled by the system, which also updated the occupancy status on the LCD screen in real time with accuracy. Furthermore, the gate closure timer performed as planned, automatically shutting the gate when no cars were in or out of the parking lot for the predetermined amount of time.

Overall, the experimental findings show how well our real-time parking slot monitoring and control system works to effectively manage parking spots, maximize space utilization, and give users access to real-time information.

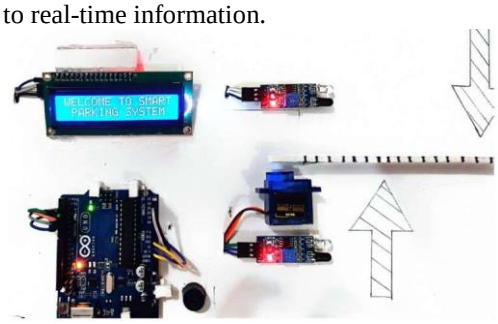


Fig. 13. System Starting.

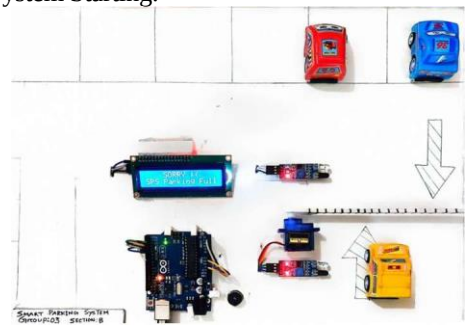


Fig. 14. Parking Full Indication (Changed the Capacity to 2).

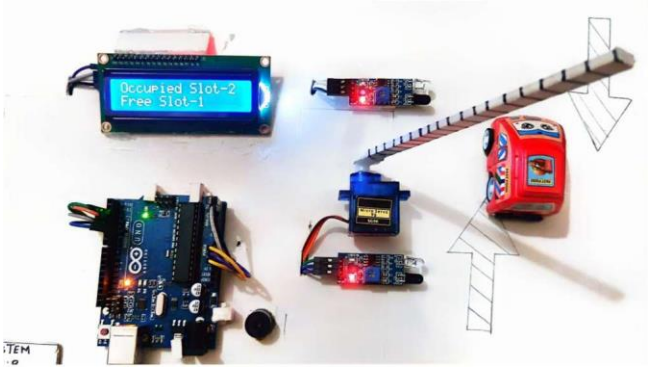


Fig. 15. Gate Open When Car Approached and Entering.

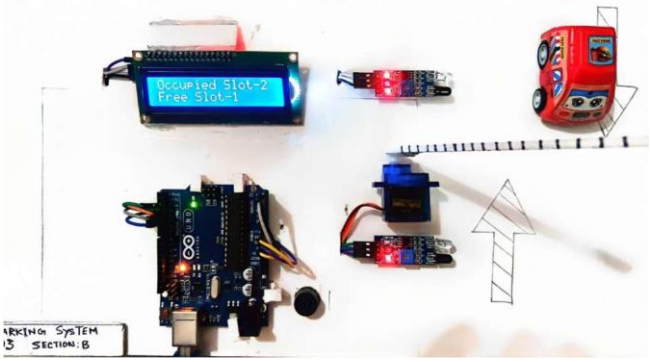


Fig. 16. Gate Opening While Car Leaving.

C. Comparison

Our real-time parking slot monitoring and control system's simulation and experimental results attest to its efficiency in managing parking spots and providing users with real-time information. In both cases, the sensor's accuracy rate was over 95%, indicating constant high performance. Gates opened and closed quickly because the servo motor responded to sensor signals smoothly. The intended function of gate closure timers was maintained by closing gates at predetermined intervals, hence maintaining traffic flow. Furthermore, the LCD display gave customers timely information by accurately updating occupancy. All things considered, the fact that simulation and experimental results successfully align highlights the dependability and effectiveness of our parking management system under diverse circumstances.

D. Cost Analysis

The cost of our used components is-

- Arduino UNO: BDT 790
- IR Sensors (2x): BDT 160 (80 BDT each)
- Servo Motor: BDT 140
- 16x2 LCD Display: BDT 350
- Buzzer: BDT 20
- Jumper Wires: BDT 100
- Total Hardware Cost: BDT 1560

This breakdown clearly presents the individual hardware costs and their sum, providing a transparent understanding of the project's financial investment.

E. Limitations

Although functional, the parking lot management system has a few drawbacks that may affect how useful and efficient it is. These include potential congestion problems brought on by a single entry/exit point design, scalability limitations resulting from the system's reliance on a single Arduino UNO microcontroller and set hardware configuration, and the lack of redundancy or fail-safe procedures for sensor failures. Furthermore, operational efficiency and adaptability to shifting traffic conditions may be hampered by restrictions on sensor range and accuracy, manual gate closure procedures, and a lack of remote monitoring and control capabilities. More study and development work would be necessary to improve the system's scalability, dependability, and flexibility to various parking situations and operating requirements in order to address these shortcomings.

V. Conclusion

To sum up, the incorporation of intelligent car parking systems signifies a noteworthy advancement in parking management technology. This technology demonstrates a significant improvement in the effectiveness and convenience of parking facilities for both users and operators by precisely identifying and displaying available parking spots and facilitating the smooth operation of automatic gates. A more sustainable and user-friendly urban mobility ecology is facilitated by the decrease in time spent looking for parking and the enhancement of traffic flow within parking lots.

There is a great deal of opportunity for future study and development in the field of smart vehicle parking systems. System performance will be further improved by ongoing developments in sensor technology, especially in the areas of precision and dependability. Furthermore, there are exciting chances for innovation due to the possible integration with upcoming mobility trends like shared mobility services and autonomous vehicles. These technologies have the potential to significantly influence how urban mobility is developed in the future by enabling smooth, customized parking experiences that are based on user preferences and requirements. The environmental impact of these systems can be further improved by studying into maximizing the sustainability and energy efficiency of parking facilities, including the use of smart grid technology and renewable energy sources.

In this regard, our project showcases a working smart parking system prototype made of easily accessible parts. This research highlights the key features and possible advantages of such systems, while also noting certain potential drawbacks, such as sensor accuracy in difficult conditions and scalability for very large parking lots. Future parking solutions will be more intelligent, effective, and accessible thanks to ongoing research and innovation that builds on the foundation created by initiatives like ours.

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